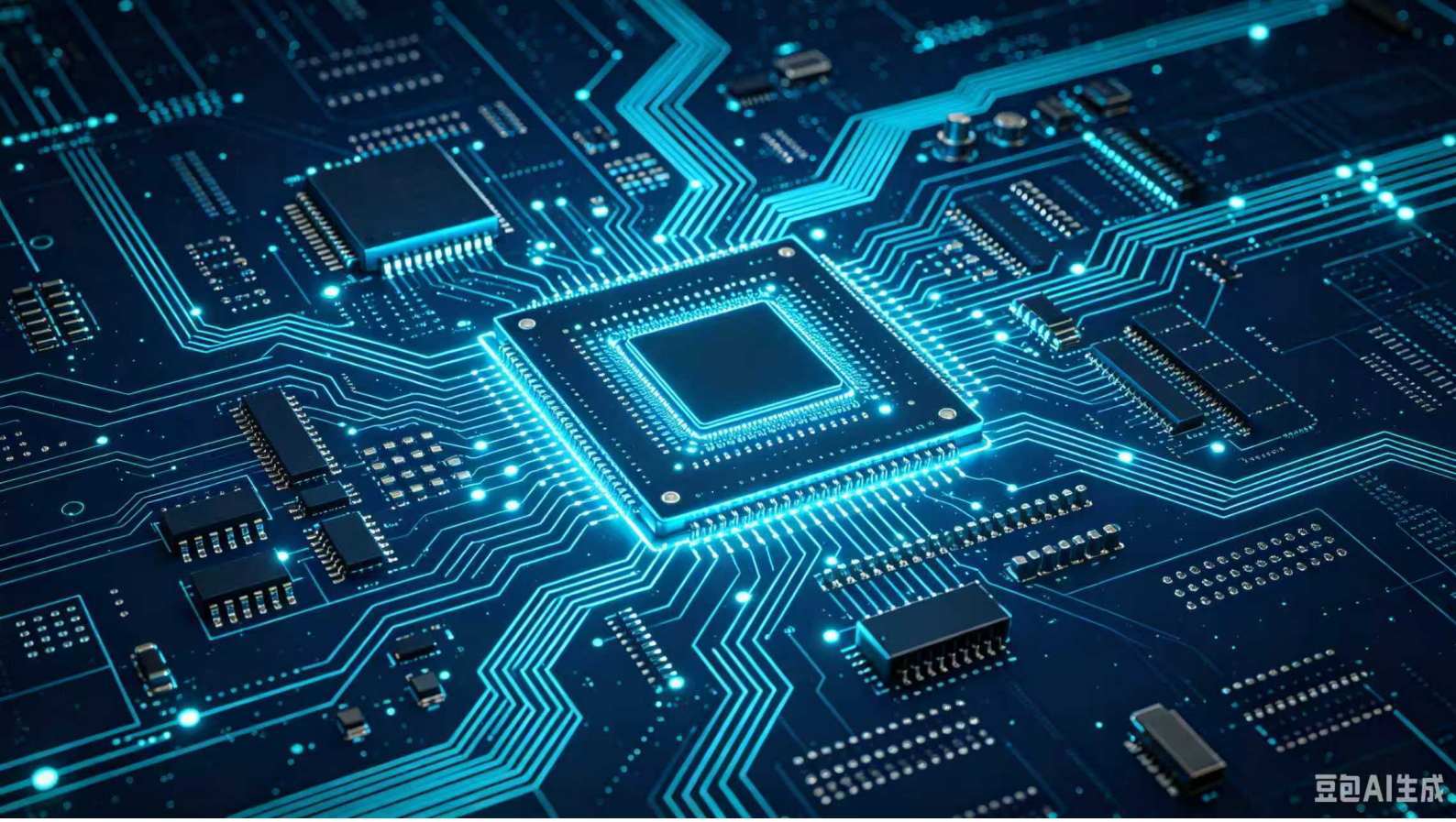


10. Magnetic Properties of Materials

> The first principle is that you must not fool yourself—and you are the easiest person to fool.
> – *Richard P. Feynman*

10.1 The Magnetic Dipole

A **magnetic dipole** is a fundamental concept in electromagnetism, representing a system that produces a magnetic field similar to that of a small bar magnet. Magnetic dipoles can arise from various sources, including:

- A current-carrying coil or loop,
- A permanent magnet,
- A rotating sphere of charge,
- Subatomic particles, such as electrons and protons.

Although a complete understanding of magnetism at the atomic level requires quantum mechanics, many important magnetic phenomena can be explained by modeling magnetic dipoles as

current loops. This classical approach provides significant insight into the behavior of magnetic materials and devices.

To characterize a magnetic dipole, we introduce the **magnetic dipole moment** $\boldsymbol{\mu}$, the same quantity that was defined for a current loop in Section 8.3section.8.3. For a planar coil with N turns, each carrying a current i, and with area vector $\mathbf{A}$ (whose magnitude is the area of the coil and whose direction is given by the right-hand rule), the magnetic dipole moment is defined as

$$\boldsymbol{\mu} = Ni\mathbf{A}. \tag{10.1}$$

The direction of $\boldsymbol{\mu}$ is perpendicular to the plane of the coil, in the direction given by curling the fingers of the right hand in the direction of current flow; the thumb then points along $\boldsymbol{\mu}$.

When a magnetic dipole is placed in an external magnetic field $\mathbf{B}$, it experiences a torque that tends to align $\boldsymbol{\mu}$ with $\mathbf{B}$. The torque is given by the vector cross product:

$$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}. \tag{10.2}$$

If θ is the angle between $\boldsymbol{\mu}$ and $\mathbf{B}$, the magnitude of the torque is

$$\tau = \mu B \sin\theta. \tag{10.3}$$

This torque can be derived from a potential energy function. The potential energy of a magnetic dipole in a magnetic field is

$$U_B = -\boldsymbol{\mu} \cdot \mathbf{B} = -\mu B \cos\theta. \tag{10.4}$$

The negative sign indicates that the system tends to minimize its energy by aligning $\boldsymbol{\mu}$ with $\mathbf{B}$.

It is important to note that, in this classical treatment, we assume the current in the loop is constant, so the magnitude of the dipole moment is fixed. The energy calculated here is the energy associated with the orientation of the dipole in the field, not the energy required to maintain the current.

Comparison with electric dipole. The magnetic dipole is fully analogous to the electric dipole:

$$\begin{aligned} \boldsymbol{\mu} &= Ni\mathbf{A}, & \mathbf{p} &= q\mathbf{d}, \\ \boldsymbol{\tau}_B &= \boldsymbol{\mu} \times \mathbf{B}, & \boldsymbol{\tau}_E &= \mathbf{p} \times \mathbf{E}, \\ U_B &= -\boldsymbol{\mu} \cdot \mathbf{B}, & U_E &= -\mathbf{p} \cdot \mathbf{E}. \end{aligned}$$

Along the dipole axis at large distances, the far-field expressions are:

$$\mathbf{B} = \frac{\mu_0}{2\pi}\frac{\boldsymbol{\mu}}{|z|^3}, \qquad \mathbf{E} = \frac{1}{2\pi\epsilon_0}\frac{\mathbf{p}}{|z|^3}. \tag{10.5}$$

The modulus is needed because both fields point along their dipole moments on *both* sides of the dipole: the exact on-axis field of a ring, $B_z = \mu_0 i a^2/2(a^2+z^2)^{3/2}$ of Eq. (9.1equation.9.1), is an even function of z and positive for every z.

10.2 Magnetic Materials

A magnetic material contains a large number of atomic or molecular magnetic dipole moments. Each dipole has a north and a south pole, and their collective behavior gives rise to the macroscopic magnetic properties of materials. The poles cannot be separated: $\nabla \cdot \mathbf{B} = 0$ (Chapter 9chapter.9) says that there is no isolated magnetic charge, so a dipole is the closest a material can come to one. The response of a material to an external magnetic field, as well as the interactions among the dipoles themselves, leads to a wide variety of magnetic phenomena.

Depending on the origin and interaction of these atomic dipole moments, materials show several kinds of magnetic behavior, including *paramagnetism*, *diamagnetism*, and *ferromagnetism*.

10.2.1 Spin Glasses and Frustration

A **spin glass** is a disordered magnetic system where magnetic atoms (e.g., manganese) are randomly mixed into a nonmagnetic host (e.g., copper). Each magnetic atom behaves as a small dipole, but the random positions lead to **frustration**: the interaction between two moments mediated by the host electrons alternates in sign as their separation grows, so random positions mean interactions of conflicting signs, and not all neighboring dipoles can simultaneously satisfy their preferred alignments. The study of spin glasses, for which Giorgio Parisi was awarded the 2021 Nobel Prize in Physics, has influenced fields far beyond physics, including mathematics, biology, neuroscience, and machine learning.

10.3 Paramagnetism

Paramagnetic materials are characterized by the presence of permanent atomic or molecular magnetic dipole moments. In the absence of an external magnetic field, these dipoles are randomly oriented due to thermal motion, resulting in zero net magnetization $\mathbf{M}$ (defined below).

When an external magnetic field $\mathbf{B}_{\text{ext}}$ is applied, the dipoles tend to align with the field, producing a net magnetization $\mathbf{M}$ in the direction of $\mathbf{B}_{\text{ext}}$. The degree of alignment depends on both the strength of the field and the temperature of the material.

The **magnetization M** of a material is defined as the net magnetic dipole moment per unit volume:

$$\mathbf{M} = \frac{\text{Net magnetic dipole moment}}{\text{Volume}} \tag{10.6}$$

Experimentally, Pierre Curie found that for many paramagnetic materials, the magnetization obeys **Curie's law**:

$$M = C\frac{B_{\text{ext}}}{T} \tag{10.7}$$

where C is the Curie constant, B_{ext} is the magnitude of the external magnetic field, and T is the absolute temperature in kelvins. Note that C is defined here through M and B_{ext} and not through χ and H, so its value differs by a factor μ_0 from the constant tabulated in that convention. This law shows that increasing the external field increases M, while increasing the temperature decreases M due to greater thermal agitation.

Curie's law is an approximation, valid when $\mu B_{\text{ext}} \ll k_B T$. For very strong fields or very low temperatures, the magnetization approaches a maximum value as all dipoles become aligned with the field:

$$M_{\max} = \frac{N\mu}{V} \tag{10.8}$$

where N is the number of dipoles (atoms) in volume V, each with magnetic moment μ.

Consider a paramagnetic solid with n atoms per unit volume, each atom possessing a magnetic dipole moment $\boldsymbol{\mu}$, so that $N = nV$ atoms occupy the volume V of the saturation formula above. The energy of a dipole in an external field is $U = -\boldsymbol{\mu} \cdot \mathbf{B}$.

If, for simplicity, the dipole can only be parallel or antiparallel to the field (as for a single electron spin), the fraction of dipoles aligned parallel to $\mathbf{B}$ is proportional to $e^{\mu B/k_B T}$, and the fraction aligned antiparallel is proportional to $e^{-\mu B/k_B T}$, where k_B is Boltzmann's constant. For $\mu B \ll k_B T$, the expansion of the two exponentials gives a net magnetization proportional to B/T, consistent with Curie's law, Eq. (10.7equation.10.7); the same model also shows the saturation of Eq. (10.8equation.10.8) as $\mu B \gg k_B T$ (Fig. 10.1figure.10.1).

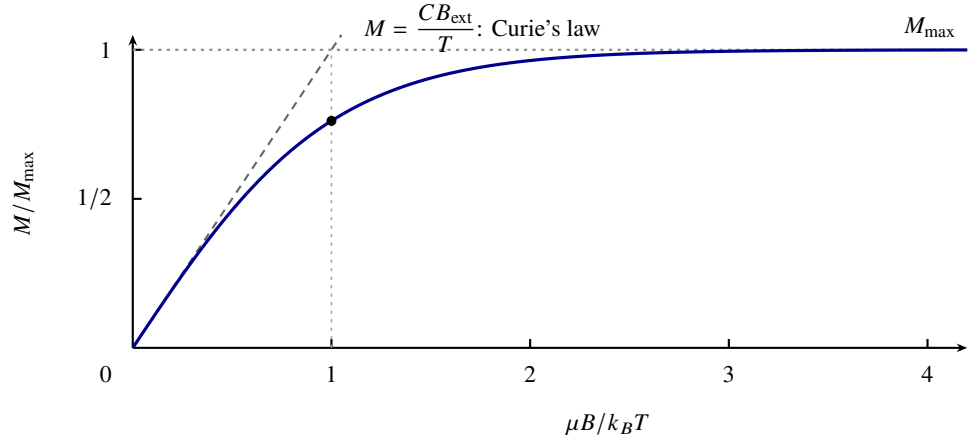


Figure 10.1: The two-state model of Section 10.3section.10.3. The solid curve is $M/M_{\max} = \tanh(\mu B/k_B T)$, the net alignment of dipoles that can only be parallel or antiparallel to the field. The dashed line is its tangent at the origin, $M/M_{\max} = \mu B/k_B T$, which is Curie's law $M = CB_{\text{ext}}/T$; the dotted line is the saturation value $M_{\max}$. At $\mu B = k_B T$, where the vertical guide is drawn, the tangent overestimates the magnetization by 31%: Curie's law is the small $\mu B/k_B T$ limit of the model.

10.4 Diamagnetism

Diamagnetism is a universal property of all materials, though it is usually masked by stronger paramagnetic or ferromagnetic effects. Diamagnetic materials are characterized by the absence of permanent atomic dipole moments. Instead, their response to an external magnetic field arises from induced currents within the atoms.

When a material is placed in a magnetic field, the field alters the motion of electrons in their orbits, generating small induced magnetic dipole moments that oppose the applied field. This effect is generally very weak and leads to a repulsion from regions of strong magnetic field.

In a simple model, diamagnetism can be viewed as arising from equal numbers of electrons orbiting in opposite directions. The external field modifies these orbits, resulting in a net magnetic moment opposite to the field direction, regardless of the direction of electron motion. To see where the sign comes from, let an electron of mass m circle a fixed center at radius r with angular speed ω, so that $m\omega^2 r$ equals the Coulomb force. In a field $\mathbf{B}$ perpendicular to the orbit the magnetic force contributes $\pm e\omega r B$, the sign depending on the sense of circulation; carrying that sense in the sign of ω itself, the force balance reads

$$m\omega^2 r = m\omega_0^2 r + e\omega r B,$$

where ω_0 is the angular speed in the absence of the field. Writing $\omega = \omega_0 + \Delta\omega$ at fixed r and keeping terms to first order in B gives $2m\omega_0\Delta\omega = e\omega_0 B$, that is $\Delta\omega = eB/2m$ — the same shift for either sign of ω_0, since ω_0 cancels. The orbit carries a magnetic moment $\mu = -\frac{e}{2}r^2\omega$, so the field induces

$$\Delta\mu = -\frac{e}{2}r^2\Delta\omega = -\frac{e^2 r^2}{4m}B, \tag{10.9}$$

opposite to $\mathbf{B}$ for either sense of motion, and proportional to the field, which is why the susceptibility of Eq. (10.12equation.10.12) is negative.

10.5 Ferromagnetism

Ferromagnetic materials, such as iron, cobalt, nickel, gadolinium, and certain alloys, exhibit strong and permanent magnetism. The distinguishing feature of ferromagnets is the presence of strong interactions between neighboring atomic dipole moments, which tend to align the moments parallel to each other, even in the absence of an external field.

The total energy of a system of interacting dipoles in an external field can be modeled as

$$U = -J \sum_{\langle ij \rangle} \boldsymbol{\mu}_i \cdot \boldsymbol{\mu}_j - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B} \tag{10.10}$$

where J measures the strength of the exchange interaction, and the first sum runs over neighboring pairs of dipoles. With the moments written as $\boldsymbol{\mu}$ rather than as dimensionless spins, J is not itself an energy: the exchange energy of a pair is $J\mu^2$. This interaction leads to a spontaneous alignment of dipoles, resulting in a net magnetization.

10.5.1 Hysteresis

A key property of ferromagnetic materials is **hysteresis**. When an external magnetic field is applied and then removed, the magnetization does not retrace its original path; instead, it follows a loop known as the *hysteresis loop*. At certain points, even with no external field, the material remains magnetized. This phenomenon underlies the permanent magnetism observed in many materials and is crucial for the storage of information in magnetic media.

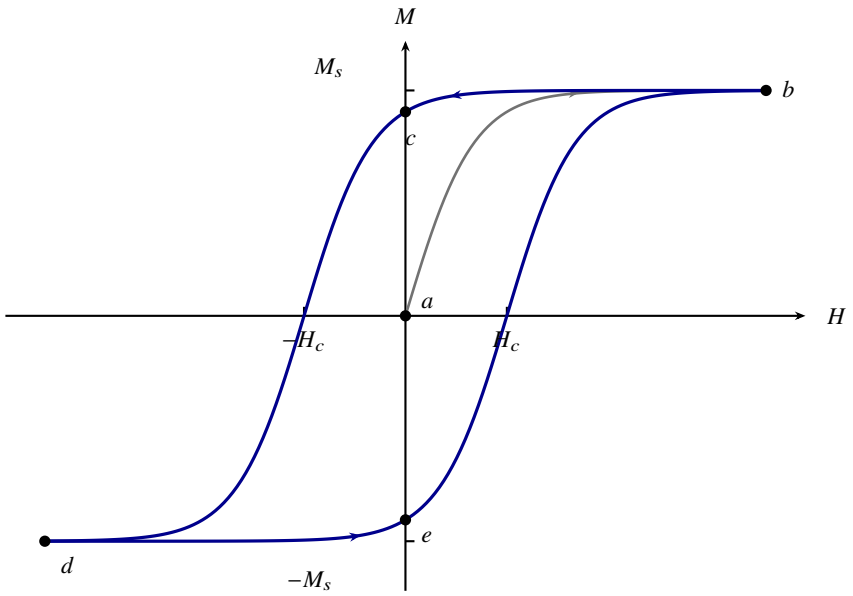


Figure 10.2: A hysteresis loop, drawn as the magnetization M against the field strength H. Starting from an unmagnetized state a, the magnetization follows the initial curve ab as the external field increases; the grey curve is that initial magnetization, and the blue loop is traced afterwards. Reducing the field to zero leaves the remanent magnetization c; driving the field further in the reverse direction carries the state through c to the negative extreme d. Bringing the field back up carries it through e, the remanence point of the reversed branch, and then to b, closing the loop b-c-d-e-b. The loop crosses $M = 0$ at the coercive field $\pm H_c$ and saturates at $\pm M_s$. It does not retrace itself, and that is the defining feature of hysteresis.

10.5.2 Magnetic Domains

Despite the strong interactions that favor alignment, not every piece of ferromagnetic material is a strong magnet in its natural state. This is because the material is composed of many small regions called **magnetic domains** (Fig. 10.3figure.10.3), within which the dipoles are aligned. The domains themselves are oriented such that their external magnetic effects largely cancel. When a strong external field is applied, these domains can grow and reorient, leading to a net magnetization of the entire specimen.

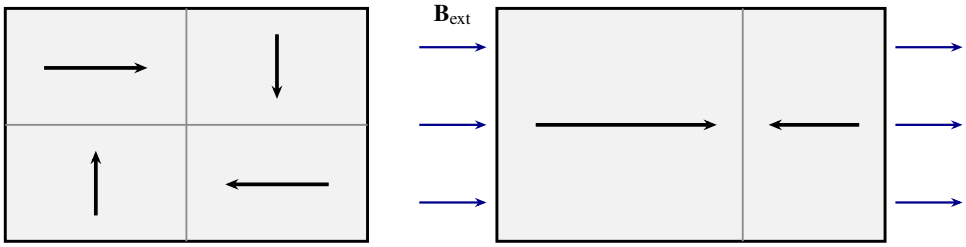


Figure 10.3: Ferromagnetic domains, shown schematically. (a) In the natural, unmagnetized state the moments of neighboring domains close on one another, so that the flux stays inside the material and its external magnetic effect largely cancels. (b) In an applied field $\mathbf{B}_{\text{ext}}$ the domains whose moments lie along the field grow at the expense of those opposed to it, and the specimen acquires a net magnetization.

10.5.3 The Curie Temperature

Thus, whether a material with permanent magnetic dipole moments is ferromagnetic, paramagnetic, or neither is decided by the competition between the exchange interaction and thermal agitation. When the exchange interaction favors parallel alignment on every bond, the alignment survives only while $k_B T$ stays below the exchange energy, of order $J\mu^2$, and the two become comparable at the **Curie temperature** T_C: below T_C such a material is ferromagnetic, while above it thermal motion defeats the exchange interaction and the same permanent dipoles are merely paramagnetic, with a susceptibility that falls off as $1/(T - T_C)$ rather than as the $1/T$ of Curie's law in Section 10.3section.10.3. Interactions of mixed sign, as in the spin glass of Section 10.2.1subsection.10.2.1, give neither: the moments are permanent and the interactions strong, yet no single alignment satisfies them all.

10.5.4 Anomalous Hall Effect

In ferromagnetic materials, an additional contribution to the Hall resistivity arises from the material's spontaneous magnetization, known as the **anomalous Hall effect**. Unlike the ordinary Hall effect, this contribution is proportional to the magnetization rather than the external magnetic field. In certain topological materials, the **quantum anomalous Hall effect** can be observed, where the Hall conductance is quantized even in the absence of an external magnetic field (Chang et al., Science 340, 167, 2013).

10.6 Magnetic Field Strength and Bound Currents

We have described a magnetic material by its magnetization $\mathbf{M}$, but we have not yet written down how $\mathbf{M}$ responds to the field applied to it. To do so, it is convenient to split the total magnetic field

B into the part produced by the free currents we control and the part produced by the material itself. This motivates the definition of the **magnetic field strength**

$$\mathbf{H} \equiv \frac{\mathbf{B}}{\mu_0} - \mathbf{M}, \qquad \text{that is} \qquad \mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}). \tag{10.11}$$

Like **M**, the field strength **H** is measured in amperes per meter (A/m); only **B** is measured in tesla.

For paramagnets with $\mu B \ll k_B T$, for diamagnets, and for ferromagnets along the initial magnetization curve of a demagnetized specimen in weak fields, the magnetization is proportional to the field strength:

$$\mathbf{M} = \chi \mathbf{H} \tag{10.12}$$

where the dimensionless constant χ is the **magnetic susceptibility**. Then

$$\mathbf{B} = \mu_0(1 + \chi)\mathbf{H} = \mu_r \mu_0 \mathbf{H} = \mu \mathbf{H} \tag{10.13}$$

which defines the **relative permeability** $\mu_r = 1 + \chi$ and the **permeability** $\mu = \mu_r \mu_0$ of the material. (The same letter μ is used for the magnetic dipole moment; which is meant will always be clear from the context, and the permeability never appears in boldface.) For a paramagnet χ is small and positive, because the permanent dipoles tend to align with the field (Section 10.3section.10.3); for a diamagnet χ is small and negative, because the induced moments oppose it (Section 10.4section.10.4); and for a ferromagnet μ_r can reach several thousand, but **M** is not proportional to **H**, and the hysteresis loop of Fig. 10.2figure.10.2 is a plot of the magnetization M against the field strength H.

The usefulness of **H** is that only the free current enters Ampère's law written for it:

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = i_{\text{free}}, \qquad \nabla \times \mathbf{H} = \mathbf{j}_{\text{free}}, \tag{10.14}$$

where i_{free} is the free current enclosed by the closed path C. These are the magnetic counterparts of $\oint_S \mathbf{D} \cdot d\mathbf{S} = q_{\text{free}}$ and $\nabla \cdot \mathbf{D} = \rho$ of Section 6.4section.6.4: just as the flux of **D** through a closed surface is fixed by the free charge it encloses, the circulation of **H** round a closed path is fixed by the free current it links; when the symmetry of the problem fixes the direction of **H** and makes its magnitude constant along such a path, this determines **H** itself, and for a linear material the medium then enters through $\mathbf{B} = \mu \mathbf{H}$ of Eq. (10.13equation.10.13).

Equivalently, the magnetization may be viewed as carrying currents of its own. A magnetized material is equivalent to a bound volume current density and a bound surface current density

$$\mathbf{j}_b = \nabla \times \mathbf{M}, \qquad \mathbf{K}_b = \mathbf{M} \times \hat{n}, \tag{10.15}$$

where $\hat{n}$ is the outward normal at the surface of the material. The total current density is $\mathbf{j} = \mathbf{j}_{\text{free}} + \mathbf{j}_b$, and the two forms of Ampère's law are consistent because $\oint_C \mathbf{H} \cdot d\mathbf{l} = i_{\text{free}}$ is the same statement as $\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0(i_{\text{free}} + i_b)$.

The full correspondence with the dielectric case of Section 6.4section.6.4, where the bound charge densities of the last row are defined, is collected below. The correspondence is not a literal substitution: **D** is built by *adding* **P** to $\epsilon_0 \mathbf{E}$, while **H** is built by *subtracting* **M** from $\mathbf{B}/\mu_0$, and the bound charge of the last row carries a minus sign where the bound current does not.

Electric	Magnetic
E (total field)	**B** (total field)
P (polarization)	**M** (magnetization)
$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$	$\mathbf{H} = \mathbf{B}/\mu_0 - \mathbf{M}$
$\oint_S \mathbf{D} \cdot d\mathbf{S} = q_{\text{free}}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = i_{\text{free}}$
$\nabla \cdot \mathbf{D} = \rho_{\text{free}}$	$\nabla \times \mathbf{H} = \mathbf{j}_{\text{free}}$
$\epsilon = \epsilon_r \epsilon_0$, $\mathbf{D} = \epsilon \mathbf{E}$	$\mu = \mu_r \mu_0$, $\mathbf{B} = \mu \mathbf{H}$
$\rho_b = -\nabla \cdot \mathbf{P}$, $\sigma_b = \mathbf{P} \cdot \hat{n}$	$\mathbf{j}_b = \nabla \times \mathbf{M}$, $\mathbf{K}_b = \mathbf{M} \times \hat{n}$

Both entries of the second-to-last row hold for a linear material only; for a ferromagnet $\mathbf{B} = \mu\mathbf{H}$ fails, exactly as $\mathbf{D} = \epsilon\mathbf{E}$ fails for a nonlinear dielectric.

10.7 Exercises

1. Consider a circular ring of radius R placed in the xy-plane and its center located at the origin. The ring carries a current I, flowing counterclockwise when viewed from the positive z axis.

   (a) For a point $\mathbf{r} = (r\sin\theta, 0, r\cos\theta)$ located in the xz-plane, show that for $r \gg R$ the magnetic field has the following form:

   $$\mathbf{B} = -\frac{\mu_0 \boldsymbol{\mu}}{4\pi r^3} + \frac{3\mu_0(\boldsymbol{\mu}\cdot\mathbf{r})\mathbf{r}}{4\pi r^5},$$

   where the magnetic dipole moment is

   $$\boldsymbol{\mu} = I\pi R^2 \hat{z}.$$

   (b) What is the potential energy of a magnetic dipole $\boldsymbol{\mu}'$ if it is placed at $\mathbf{r}$ ($r \gg R$)?

2. A long solenoid with n turns per unit length carries a free current i, and its interior is completely filled with a material of relative permeability μ_r. Find the magnetic field strength H, the magnetic field B, the magnetization M, and the bound surface current K_b on the surface of the material.

3. A paramagnetic solid contains n identical atoms per unit volume, each carrying a permanent magnetic dipole moment of magnitude μ. The solid is held at absolute temperature T in a uniform external magnetic field $\mathbf{B} = B\hat{z}$ with $B > 0$, and the dipoles do not interact with one another. Assume, as in Section 10.3, that each moment can point only parallel or antiparallel to $\mathbf{B}$, and that the fractions of atoms in the two states are proportional to $e^{\mu B/k_B T}$ and $e^{-\mu B/k_B T}$ respectively, where k_B is Boltzmann's constant.

   (a) Find the magnetization M of the solid. Show that for $\mu B \ll k_B T$ it reduces to Curie's law $M = CB/T$, and give both the Curie constant C and the saturation magnetization $M_{\max}$ in terms of n, μ and k_B.

   (b) Find the value of B/T at which the magnetization reaches half of $M_{\max}$.

4. In a classical model of diamagnetism, an electron of charge $-e$ (with $e > 0$) and mass m moves in a circle of radius r about a fixed nucleus, the electrostatic attraction supplying the centripetal force, with angular speed ω_0. A uniform magnetic field $\mathbf{B} = B\hat{n}$ with $B > 0$ is then applied along the normal $\hat{n}$ to the plane of the orbit, slowly enough that the electron stays on a circular orbit, of the same radius r, at every stage. Work to first order in B.

   (a) Find the new angular speed, for each of the two senses in which the electron may circulate about $\hat{n}$.

   (b) Find the change $\Delta\boldsymbol{\mu}$ in the orbital magnetic dipole moment in each case, and show that $\Delta\boldsymbol{\mu}$ opposes $\mathbf{B}$ whichever way the electron circulates.